Learning with Less: Comparing Word and Character Features for News Classification in Four Nigerian Languages

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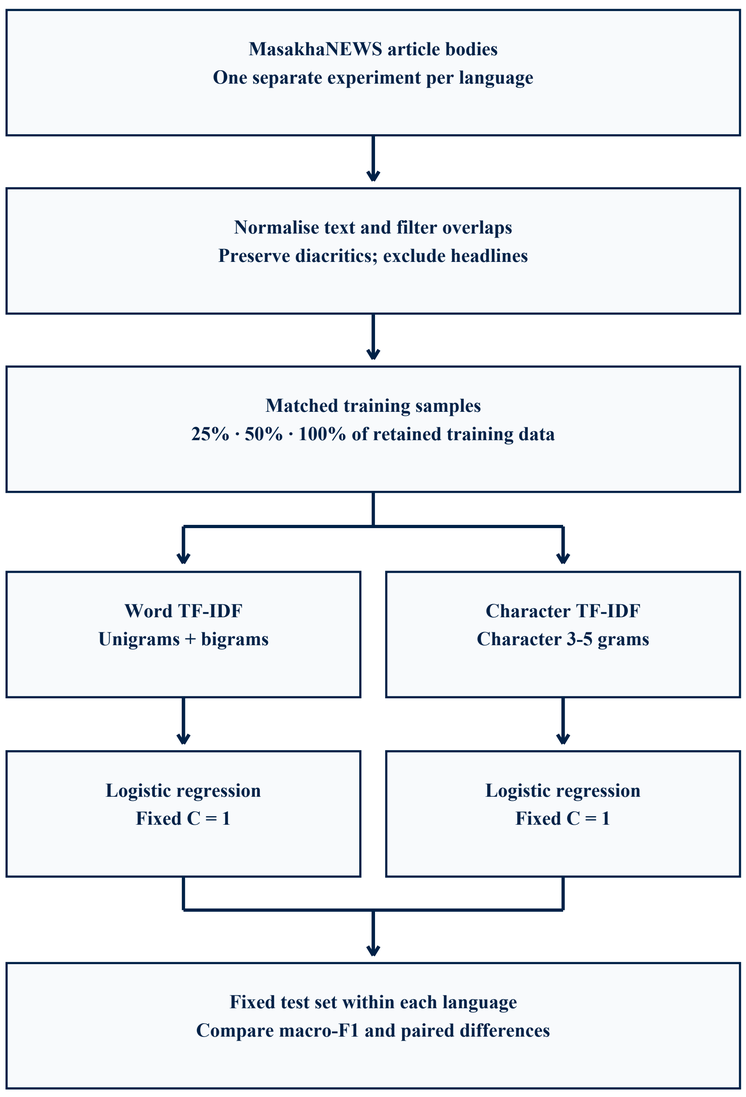
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Character features scored higher in all 12 comparisons.
Full-data differences remain uncertain.

Introduction

- **Question:** How do word and character features compare as the amount of labelled training data changes?
- **Scope:** MasakhaNEWS article bodies in Yorùbá, Igbo, Hausa and Nigerian Pidgin.
- **Purpose:** evaluate inexpensive classical baselines before more complex modelling.

Materials and Methods



Feature Extraction Example: “Sapa choke”

Word TF-IDF · Unigrams + bigrams

[“Sapa”, “choke”, “Sapa choke”]

Character TF-IDF · 3-5 grams

[“Sap”, “apa”, “pa·”, “a·c”, “·ch”, “cho”, “hok”, “oke”, “Sapa”, “apa·”, “pa·c”, “a·ch”, “·cho”, “chok”, “hoke”, “Sapa·”, “apa·c”, “pa·ch”, “a·cho”, “·chok”, “choke”]

· marks a space. Case shown before lowercasing.
Illustration only; not a dataset example or fitted TF-IDF weights.

Preprocessing and evaluation

- **Normalisation:** NFC, lowercase and whitespace cleanup. Unicode word tokenisation retains diacritics.
- **Overlap filtering:** exact text, repeated URL, or ≥0.80 Jaccard similarity on word 5-grams. First-retained priority: train → dev → test.
- **Sampling:** five matched, stratified subsets at 25% and 50%; each full-data model fitted once.
- **Training only:** learn TF-IDF vocabulary and weights from the corresponding training sample; maximum 50,000 features.
- **Uncertainty:** 10,000 paired test-article bootstrap resamples; 95% percentile intervals for Character minus Word.

Data retained after overlap filtering

Language	Train	Test	Classes
Yorùbá	1,433	411	5
Igbo	1,295	361	6
Hausa	2,216	633	7
Nigerian Pidgin	1,058	300	5

125 rows removed across train, dev and test. Removed rows are not necessarily independent stories.

Results and Discussion

Full-data scores

Macro-F1 (0-100); higher is better.

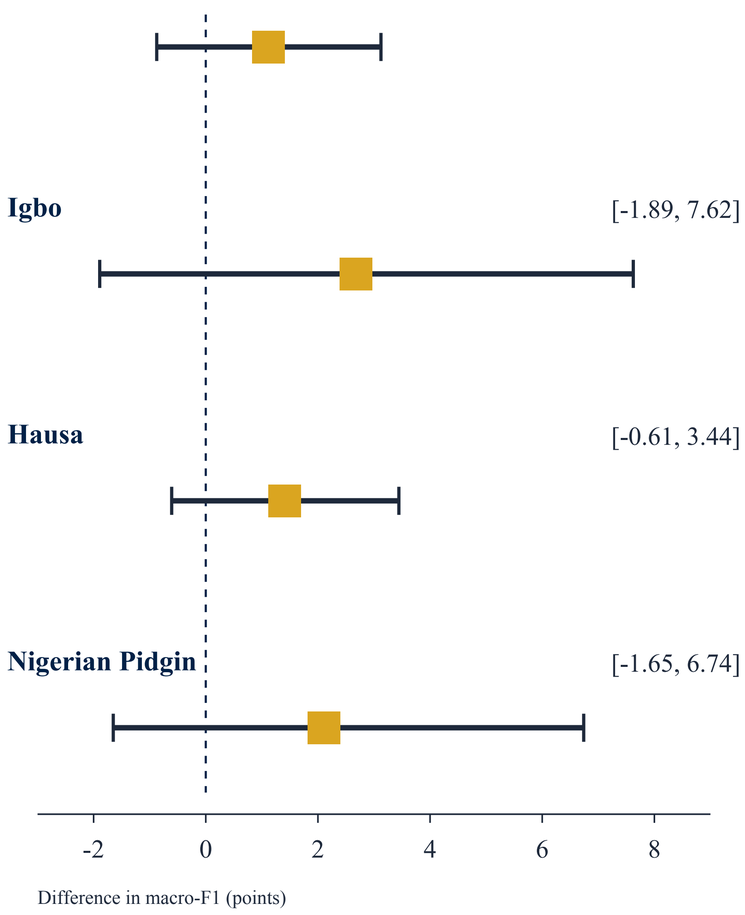
Language	Word	Character	Difference
Yorùbá	86.79	**87.90**	+1.11
Igbo	69.94	**72.63**	+2.68
Hausa	85.26	**86.67**	+1.41
Nigerian Pidgin	82.75	**84.86**	+2.11

Differences are calculated before rounding. Each language has a different test set and category set.

How uncertain is the advantage?

Full-data Character minus Word macro-F1
Pointwise 95% paired bootstrap intervals

All four intervals cross zero, so the full-data advantage remains uncertain.



All four intervals include zero.

The intervals capture test-sample uncertainty conditional on fitted models. They do not capture training variability or establish equivalence.

Performance across training sizes

Both methods improve with more data; character features score higher at every tested size.



Axes: macro-F1 (0-100) against training data used (%).
25% / 50%: mean ±1 sample SD over five subsets.
100%: one fit per model; no subset-SD bar.

Conclusion and Recommendations

- **Consistent direction:** character features lead in all 12 language-size comparisons (+1.11 to +3.05 points). The full-data advantage remains uncertain.
- **Data size:** both methods improve with more data; the character advantage does not consistently widen at smaller sizes.
- **Limitations:** fixed C = 1 and a shared 50,000-feature cap may affect the comparison; the cap can constrain the representations differently.
- **Per-class results:** gains are uneven; estimates for small categories are especially uncertain.
- **Next:** tune regularisation and feature limits separately, add a combined word-and-character baseline, and evaluate beyond news.

Interpretation: The 12 comparisons share data and are not independent. Differing test-set sizes and categories preclude ranking language difficulty. Filtered splits differ from published MasakhaNEWS baselines.

Source: Adelani et al. (2023), MasakhaNEWS. https://aclanthology.org/2023.ijcnlp-main.10/